

AI-Native Portfolio Guide

A Complete Technical Walkthrough for Interview Preparation (English & Hinglish)

PART 1: English Guide

1. Concept & Architectural Overview

This project reimagines the traditional personal portfolio website. Instead of forcing visitors to scroll through static text boxes, it establishes an **AI-Native Conversational Interface**. The visitor converses with an AI persona of Monit Kumar Jangir. The AI agent answers questions about Monit's experience and can **dynamically invoke rich React UI components** (such as project carousels, skills lists, and contact forms) directly inside the chat dialog using real-time function calling.

2. Technology Stack & Directory Structure

The project is structured under the Next.js framework using the following tools:

Layer	Technology Used	Role in Portfolio
Frontend Framework	Next.js 15 (App Router, TS)	Provides routing, server/client component hydration, and SEO optimization.
Styling & Motion	Tailwind CSS + Framer Motion	Tailwind manages the layout system, while Framer Motion handles smooth transitions.
AI Orchestration	Vercel AI SDK (Unified SDK)	Handles chat streams, handles tool definitions, and parses responses from the LLM.
LLM Model Provider	Mistral AI API (Mistral Large)	Powers the chat logic, understands visitor queries, and triggers tool executions.
Voice Synthesis & STT	Web Speech API (Browser Native)	Enables Speech-to-Text (microphone) and Text-to-Speech (reading aloud) for free.

3. Core Components Working

A. Home Page (page.tsx): Acts as the entry portal. It loads a 3D-fluid-particle background, a prominent thumbs-up avatar image (transparent v3 asset), a search bar, and 5 quick action buttons (Me, Projects, Skills, Fun, Contact). It includes browser-native Speech Recognition (microphone) that lets users speak their queries, transcribes them, and auto-submits them to trigger the chat routing instantly.

B. Chat Route (chat/page.tsx & chat.tsx): Contains the core chat view. It initializes the Vercel AI SDK useChat hook. The hook handles the streaming response of text message chunks from the LLM, managing message history.

C. AI Agent Tools (tools/): When the user asks a structured question (e.g. 'Show me your projects'), the LLM stops generating text and responds with a tool call. The Vercel AI SDK intercepts this tool call. On the client side, tool-renderer.tsx intercepts the tool name (e.g., getProjects) and mounts the rich UI component (e.g., <AllProjects /> carousel) in the chat window rather than showing plain text.

4. Voice Agent Implementation Details

To make it voice-enabled, we integrated the **native Web Speech API**. This runs 100% locally in the browser with zero cost. The voice input (STT) uses `webkitSpeechRecognition`, which listens, updates the input box, and submits. The voice output (TTS) uses `window.speechSynthesis`. When the LLM finish streaming (`onFinish` callback), we clean up any markdown formatting (asterisks, code blocks, headers) and speak the plain text using a selected English voice. The talking mouth movements of your video avatar are programmatically synchronized (playing when speech starts, pausing when it ends).

PART 2: Hinglish Guide (Interview Preparation)

1. Concept Aur Working Flow

Yeh portfolio ek **AI-Native Conversational Interface** par built hai. Normal resume websites par users ko experience scroll karna padta hai, par yaha visitors directly Monit ke AI Avatar se chat kar sakte hain. AI bilkul Monit ki tarah respond karta hai kyunki humne backend par ek custom system prompt (prompt.ts) inject kiya hai jo Monit ki personal aur professional background (Collegedunia, Lawxygen AI) ki sari details hold karta hai. Sabse unique baat yeh hai ki AI sirf normal text nahi balki dynamic React elements (badgeless skill UI, contact form, projects carousel) bhi chat box ke andar responsive card form mein render kar sakta hai.

2. Frontend Frameworks Aur Sizing

- **Next.js App Router (React):** Isme hum client-side component dynamic actions ke liye React hooks use karte hain aur compile and SEO speed badhane ke liye dynamic routing use karte hain.
- **Tailwind CSS:** UI layout ko responsive aur modern design देने के लिए standard Tailwind CSS styling variables use kiye hain.
- **Framer Motion:** Smooth client animations aur landing modal movement transitions ko premium custom-feel देने के लिए iska use kiya hai.
- **Web Speech API:** Speech-to-Text (Mic icon) के लिए browser का built-in webkitSpeechRecognition and Text-to-Speech (Speaker icon) के लिए speechSynthesis use किया है। Yeh bilkul free है, koi api key charge nahi lagta.

3. Backend, AI integration Aur Tool Flow

Backend logic Next.js API route `/api/chat/route.ts` के through run होता है। Vercel AI SDK aur Mistral AI के integration से हम user query aur system prompt को package करके AI model को forward करते हैं।

Tool Calling Process (Aasan shabdo mein):

1. User type karta hai: *'Show me your projects'*.
2. AI model detect karta hai ki text generation ki jagah project show karne wala action trigger karna chahiye. AI model reply mein text return karne के bajaye tool call `getProjects` execute karta है।
3. Client side par `tool-renderer.tsx` is response को intercept karta है aur static text block की jagah `<AllProjects />` custom carousel component inject कर देता है।

4. Voice Aur UI Gap Alignment Fixes

- **Avatar Transparent BG:** user-uploaded new memoji png का background, python script के help से flood-fill (magic wand logic) perform करके transparent किया गया। Edge aliasing issues को bypass करने के लिए soft-keying mapping and browser cache busting (v2/v3 renaming) implement किया गया।
- **Card Layout & Padding Alignment:** Project titles (jaise *AI-Native Conversational Portfolio*) mobile cards में overflow होकर right edge में clip हो रहे थे। Horizontal padding को 32px से कम करके 20px किया गया aur title text sizes को slightly scale-down (text-xl to text-lg) किया गया। Isse text dynamic wrapping fit हो गया।

Interview Mock Q&A; (Hinglish)

Q: Next.js use करने का क्या benefit हुआ इस project में?

Ans: Next.js hume SSR (Server-Side Rendering) aur API routes ki capabilities ek sath deta hai. `/api/chat` route par stream handles aur prompt protection secure tarike se implement ho paye aur SEO optimize raha.

Q: Vercel AI SDK aur normal chat-streaming APIs me kya difference hai?

Ans: Vercel AI SDK multi-model support standardizes karta hai. Tool calling configurations seamlessly handle ho jati hain aur frontend par hooks (jaise `useChat`) states aur render tracking code bohot chota kar dete hain.

Q: Client-side Voice functionality ka performance impact kya hai?

Ans: Web Speech API directly browser runtime engine use karta hai. Iska latency impact minimal hai kyunki audio data compile hone ke liye servers par nahi bhejna padta, jisse processing fast aur completely costless rehti hai.